

Report 3

1. Theoretical solution for the circuit

$$V_{BE} = 0.7 \text{ V}$$

$$V_{CE} = 0.2 \text{ V}$$

$$V_{RC} = V_{CC} + V_{CE(sat)} = 15 \text{ V} - 0.2 \text{ V} = 14.8 \text{ V}$$

$$I_C = 14.8 / 1 \text{ k}\Omega = 14.8 \text{ mA}$$

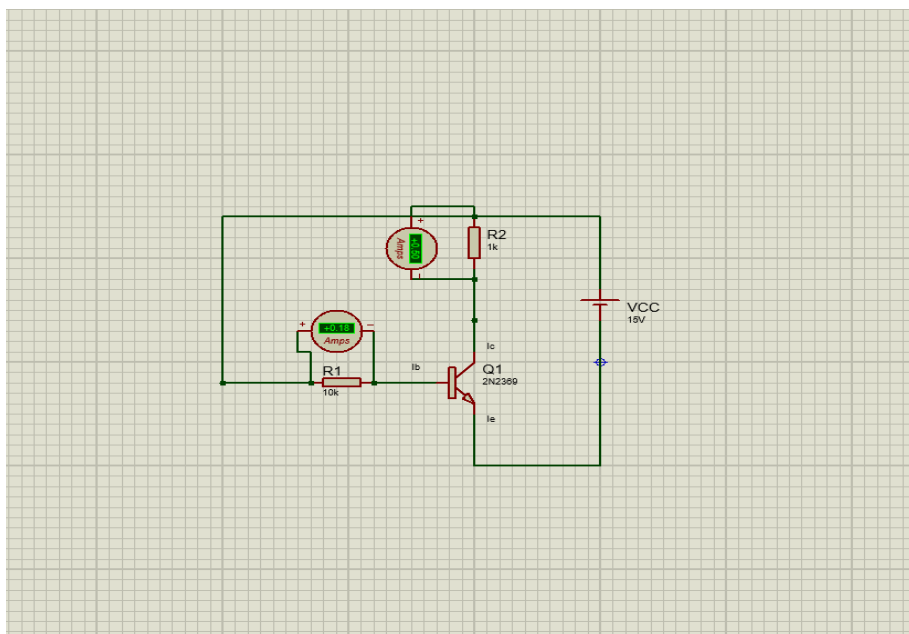
$$I_B = 14.3 / 10 \text{ k}\Omega = 1.43 \text{ mA}$$

$$I_E = 16.23 \text{ mA}$$

$$I_C / I_B = 10.35$$

2. Measurements

I_B	I_C	I_E	β
0.18	0.5	0.68	2.77



In conclusion, theoretically the values are smaller than the measurements.

Report 4

1. Theoretical solution

$$V_b = V_{CC} \cdot R_4 / (R_1 + R_4) = 12 \cdot (4.7 \text{ k}\Omega / 10 \text{ k}\Omega + 4.7) = 3.83 \text{ V}$$

$$V_e = V_b - V_{be} = 3.83 - 0.7 = 3.13 \text{ V}$$

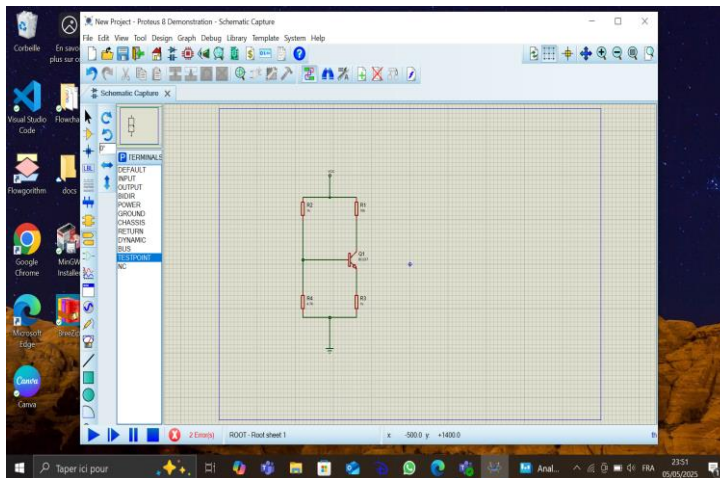
$$I_e = V_e / R_3 = 3.13 / 1 \text{ k}\Omega = 3.13 \text{ mA}$$

$$I_c = (\beta / (\beta + 1)) I_e = 120 / 121 (3.13) = 3.10 \text{ mA}$$

$$I_b = I_e - I_c = 3.13 - 3.10 = 0.026 \text{ mA} = 26 \mu\text{A}$$

$$V_C = V_{CC} - (I_C \cdot R_2) = 12 - (3.10\text{mA} \cdot 1\text{k}\Omega) = 8.90\text{V}$$

Measurements



I_B	I_C	I_E	V_B	V_C	V_E
0.16	0.01	0.12	3.83	9	3.1

In conclusion, measurements are smaller than theoretical values.

Experience 5 :

- 1) An op-amp is an integrated circuit that amplifies the voltage difference between its two inputs.

Op-amps are used in:

- Amplifiers
- Filters
- Voltage regulators

- 2) The amplification factor can be up to 100,000 times or more.

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